SQ26 Part 2 — QDArchive Classification Report

Student ID: 23071063 | Final report generated from the staging and official delivery SQLite databases

Executive Overview

This report documents the complete Part 2 workflow: peer/shared metadata ingestion, source normalization, project-type classification, duplicate analysis, Tier 1 metadata evidence, Tier 2 primary-file content evidence, ISIC Rev. 5 classification of the MY_CORE delivery scope, and release automation checks.

44	174,891	4	507
Registered source databases	Staged project records	Final ISIC-classified projects	Final classified primary files

Scope distinction

Scope	Databases	Projects	Purpose
PEER_SHARED	43	174,875	Imported, normalized, profiled, project-type classified, and included in duplicate analysis.
MY_CORE	1	16	Assigned repositories used for final Part 2 delivery and ISIC classification.

Important interpretation: peer/shared records were used to exercise the multi-source staging pipeline and project-type classification workflow. The final ISIC project and file classification scope was intentionally limited to MY_CORE. The report does not claim that every peer project received an ISIC label. The MY_CORE source database contained 16 staged records; the official final delivery materialized the 9 records belonging to the assigned repositories 5 and 15.

1. Shared Corpus Ingestion and Staging

The staging database consolidates peer-shared and MY_CORE metadata databases into a normalized data model. Every source database is registered with its origin, storage type, schema signature, checksum, file size, and import audit information.

44	1,923,231	1,428,460	481,406
Completed import runs	Imported file records	Imported keywords	Imported person-role records

Import and quality summary

Measure	Result	Interpretation
Completed imports	44	All registered source imports completed successfully.
Imported projects	174,891	Metadata project records normalized into stg_projects.
Imported licenses	155,950	License records retained for source-level provenance.
Duplicate clusters	56,164	Candidate duplicate groups detected across the multi-source staging corpus.
Duplicate memberships	119,917	Project-to-cluster memberships supporting duplicate review.
High-confidence duplicate clusters	56,164	Deterministic duplicate clusters resolved in the derived analysis layer; raw staging records remain unchanged.
Duplicate records excluded from derived analysis	63,753	High-confidence duplicate records excluded non-destructively from analysis only; each retains a canonical reference.
Projects retained in derived analysis	111,138	Records retained after duplicate exclusion for engineering analysis and evaluation.
Recorded data-quality issues	0	No unresolved staging quality issues recorded.

Duplicate-resolution interpretation: duplicate candidates are clustered using deterministic identifier and metadata evidence. All registered candidate clusters met the configured high-confidence decision rule; therefore, candidate and high-confidence cluster counts are identical. High-confidence duplicate records are excluded only from a derived analysis layer. No raw project record is deleted, overwritten, or merged; each exclusion preserves its canonical reference, evidence, confidence, tags, and audit metadata.

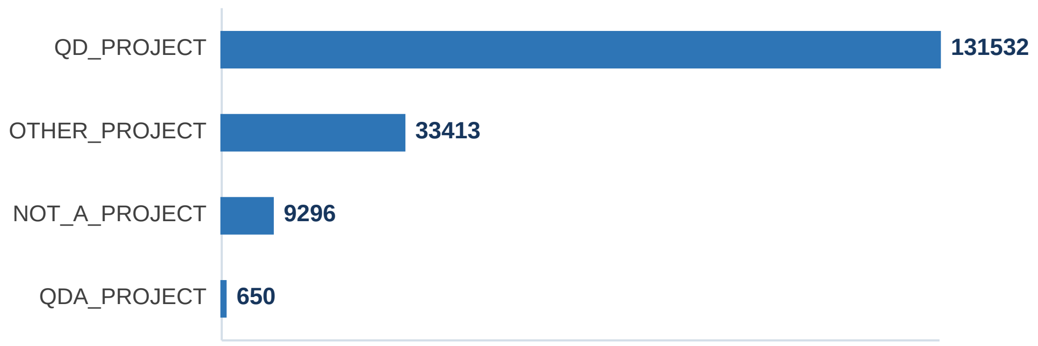
Peer/shared data contributed to the engineering workflow by providing heterogeneous schemas, mixed metadata completeness, varying repository structures, and realistic duplicate conditions. It was therefore valuable for validating that the pipeline does not assume a single repository-specific schema.

2. Project-Type Classification Across the Staging Corpus

The staging corpus was classified into QDA_PROJECT, QD_PROJECT, OTHER_PROJECT, and NOT_A_PROJECT. This stage determines which records are eligible for later qualitative-data and ISIC classification steps.

Project type	Projects	Meaning
QD_PROJECT	131,532	Project contains qualitative data evidence.
QDA_PROJECT	650	Project contains qualitative-data-analysis artefacts.
OTHER_PROJECT	33,413	Project does not provide sufficient QD/QDA evidence.
NOT_A_PROJECT	9,296	Record does not represent an eligible research project.

Project-type distribution across the staging corpus



The broad project-type stage was applied across the complete staging corpus. The final delivery, however, contains only the assigned MY_CORE repositories and their relevant project records.

3. Tier 1 and Tier 2 Classification Method

The final ISIC classification uses a transparent deterministic, rule-based evidence strategy. Tier 1 establishes project context from project titles, descriptions, repository metadata, QDPX project metadata, and keywords where available. Tier 2 uses the actual contents of primary TXT and PDF files embedded within QDPX archives. Keywords were considered where available; no keyword records were available for the final MY_CORE delivery scope.

Stage	Input evidence	Output
Source ingestion	Peer and MY_CORE SQLite databases, source registry, checksums, schema signatures, import audit.	Normalized staging records with provenance.
Project-type classification	Project metadata, file names, extensions, QDA/QD indicators, archive structure.	QDA_PROJECT, QD_PROJECT, OTHER_PROJECT, or NOT_A_PROJECT.
Tier 1 evidence	Project title, description, repository metadata, QDPX project metadata, and keywords where available.	Project-level ISIC candidate context.
Tier 2 evidence	Embedded TXT and PDF primary-file contents extracted from sources/ inside QDPX archives.	Direct file-level ISIC evidence where sufficient.
Fallback policy	Files without enough individual evidence retain the project class with documented lower confidence.	Documented project-context fallback classification.

Final MY_CORE Tier 2 evidence results

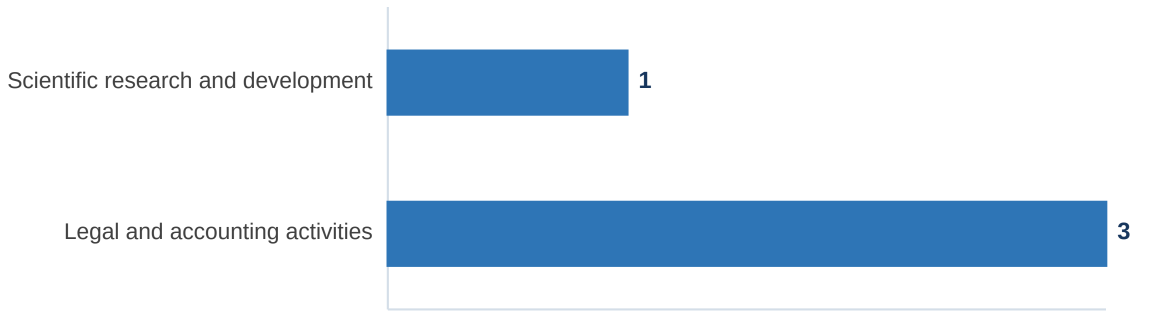
109	217	326	181
Multi-term direct files	Single-term direct files	Total direct Tier 2 files	Project-context fallback files

All 507 eligible QDPX-internal primary files in the final DANS delivery scope were processed. A direct content classification was possible for 326 files. The remaining 181 files lacked sufficient unambiguous standalone evidence and therefore used the documented project-context fallback. Raw extracted primary text was not stored in the final delivery database.

4. Final Delivery Results — Repository 5 (DANS)

4	507	326	181
QDA projects	QDPX-internal primary files	Direct Tier 2 decisions	Fallback decisions

Primary ISIC class distribution by project



Primary ISIC class distribution by primary file

Primary ISIC class distribution by primary file

Primary ISIC class ranking

Rank	Primary ISIC class	Division	Projects	Primary files
1	Legal and accounting activities	69	3	499
2	Scientific research and development	72	1	8

Top-20 clarification: only two ISIC primary classes occurred in Repository 5. Therefore, the ranking table contains two rows rather than twenty empty or artificial rows.

Interpretation

Repository 5 contains four QDA projects. Three are classified as Legal and accounting activities (ISIC division N69), covering 499 primary files. One is classified as Scientific research and development (ISIC division N72), covering 8 primary files. The direct Tier 2 evidence and fallback policy allow every primary file to remain traceable to a documented classification rule.

ISIC section summary: both final divisions, N69 and N72, belong to ISIC section M — Professional, scientific and technical activities. Therefore, all four final ISIC-classified DANS QDA projects and all 507 classified primary files belong to section M.

5. Final Delivery Results — Repository 15 (ICPSR)

5	**5**	**0**	**0**
Final project records	OTHER_PROJECT records	Primary QDA/QD files	ISIC classes assigned

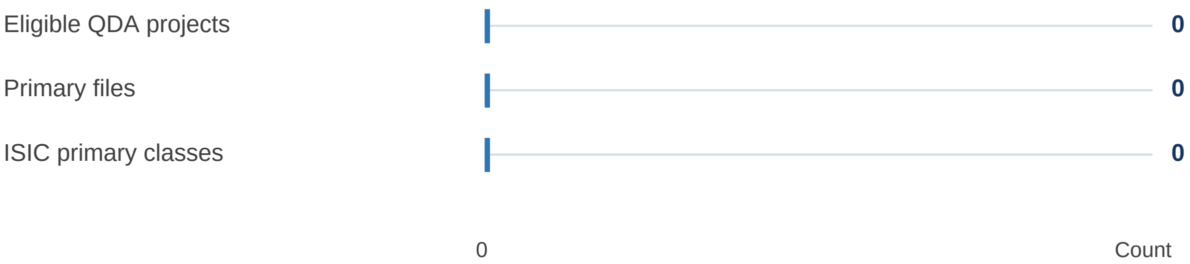
No eligible QDA/QD project was available for final ISIC classification in Repository 15.

This is an accurate repository outcome, not a missing classification result. The five retained records are metadata projects classified as OTHER_PROJECT. Because there are zero eligible QDA projects and zero primary files, a primary-class histogram and Top-20 class ranking are not applicable.

ICPSR zero-distribution histogram

Eligible QDA/QD distribution — zero observations

Each bar terminates at zero because no eligible QDA/QD project or primary file exists in Repository 15.



Interpretation

Repository 15 contributes five metadata records to the final delivery. These records do not provide QDA or QD primary-file evidence in the final delivery database. Assigning an ISIC activity class without qualifying evidence would be methodologically unsupported. Therefore, the XLSX export correctly retains empty primary_class and secondary_class fields for these records.

Repository ID	Project type	Projects	Primary files	ISIC project classes
15	OTHER_PROJECT	5	0	0

6. Automation, Monitoring, and Reproducibility

The final delivery is generated through reproducible scripts rather than manually edited data. The workflow materializes the official SQLite database, generates the XLSX/PDF deliverables, runs a read-only drift monitor, and runs a release quality gate.

Available	0	Passed	0
Drift monitor status	Projects requiring reclassification	Release quality gate	Failed quality checks

Automation component	Purpose	Observed result
Official delivery materializer	Builds 23071063-sq26-classification.db from the staging classification records.	Foreign-key and integrity validation passed.
Tier 2 extractor	Extracts bounded TXT/PDF evidence from QDPX sources/ entries without persisting raw text.	507 primary files processed.
Drift monitor	Compares database, project, and QDPX archive fingerprints without changing classifications.	Baseline stable after Tier 2 rebuild; no project required reclassification.
Release quality gate	Checks expected database structure, counts, final classification distribution, and drift status.	Passed with zero failed checks.
Automated test suite	Tests Tier 2 extraction, ISIC rules, drift monitoring, quality gate logic, and output assumptions.	46 tests passed.

Reproducibility commands

```
python -m scripts.classify_official_isic --staging-db data/staging/qdarchive_x_staging.db
--raw-data-root data/raw --reset
```

```
python -m scripts.materialize_sq26_classification_delivery
```

```
python -m scripts.run_drift_monitor
```

```
python -m scripts.run_release_quality_gate --require-drift-report
```

```
python -m scripts.generate_part2_deliverables
```

```
python -m pytest -q
```

7. Technical Data Challenges and Limitations

The following issues relate to the structure and evidential quality of the acquired data. They are documented as data handling considerations rather than programming defects.

Observed challenge	Handling and impact
Nested QDPX archive structure	DANS QDA projects were supplied as QDPX archives. Primary data were embedded in internal sources/ paths. The pipeline enumerates these entries and does not treat the outer QDPX archive itself as a primary file.
Mixed TXT and PDF sources	Primary materials contained both text files and PDFs. TXT was decoded directly; PDF text was extracted with bounded page and character limits to support classification without storing raw source text.
Incomplete standalone file evidence	Some individual files contain generic or contextual content. They cannot always support an independent ISIC decision. These files use the explicit TIER2_PROJECT_CONTEXT_FALLBACK rule.
Peer versus final ISIC scope	Peer/shared records were used for staging, project-type classification, source profiling, and duplicate analysis. ISIC labels were not assigned across the entire peer corpus; the final ISIC scope was MY_CORE only.
Non-fatal PDF encoding notices	Some PDFs produced SymbolSetEncoding notices during parsing. Extraction completed successfully for the processed corpus and classification finished with zero recorded errors.

Conclusion

The Part 2 implementation provides a traceable classification workflow over a large peer/shared staging corpus and a focused MY_CORE final delivery. The engineering extension identified and resolved 56,164 high-confidence duplicate clusters in the derived analysis layer, excluding 63,753 duplicate records from the derived analysis layer while preserving all raw staging records. The final output includes 9 delivery project records, 4 ISIC-classified QDA projects, 507 classified primary files, Tier 2 evidence provenance, reproducible report generation, drift monitoring, release validation, and a passing automated test suite.